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Restraining Device to Inhibit Removal of Footwear

Abstract

A restraining device configured to inhibit the removal of an article of footwear to which the restraining device is attached is provided. The restraining device will prevent footwear from being removed regardless of the type of fastening means used to secure the footwear to the foot of a wearer and whether the fastening means are fastened. In some embodiments, the restraining device is configured to be permanently attached to an article of footwear. In other embodiments, the restraining device is configured to be removably attached to an article of footwear. In some embodiments, the restraining device generally includes a ring-shaped member, a tightening means, and a locking mechanism. In other embodiments, the restraining device generally includes a restraining strap and a unique locking mechanism.

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Background/Summary

CROSS RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/121,148, filed Mar. 14, 2023, which is a continuation of U.S. application Ser. No. 17/517,161, filed Nov. 2, 2021, which claims priority to U.S. provisional application No. 63/110,907 filed Nov. 6, 2020, all of which are incorporated by reference herein in their entirety.

FIELD OF THE INVENTION

[0002] The present invention relates generally to a restraining device configured to inhibit the removal of footwear from a wearer of the footwear.

BACKGROUND OF THE INVENTION

[0003] Footwear is used for many purposes, including but not limited to protection, comfort, and fashion. Generally, footwear is secured to the foot of a wearer by shoelaces or hook-and-loop type systems. The shoelaces, hook-and-loop straps, or similar systems for securing the footwear to the foot of a wearer are configured to tighten the footwear to the foot of a wearer, thus preventing removal when the system is secured.

[0004] One problem with these systems of securing footwear to the foot of a wearer is that while they are easily secured, they are also easily loosened and removed. While this is desirable for some wearers of footwear, others, such as small children or adults with special needs, often remove their footwear at inopportune or inappropriate times in spite of the fact that they are fastened in place. These wearers know how to loosen the general systems for securing footwear, and thus are capable of removing the footwear at their liking. Thus, there is a need for a device configured to inhibit the removal of footwear by preventing the release of the footwear from the foot of a wearer, regardless of whether the footwear is fastened to the foot of a wearer.

SUMMARY OF THE INVENTION

[0005] A restraining device configured to prevent footwear from being removed from the foot of a wearer is provided. The restraining device will prevent footwear from being removed regardless of the type of fastening means used to secure the footwear to the foot of a wearer and whether the fastening means are fastened. Generally, the restraining device includes a ring-shaped member, a tightening means, and a locking mechanism. In some embodiments, the restraining device is configured to be permanently attached to an article of footwear. In other embodiments, the restraining device is configured to be removably attached to an article of footwear.

[0006] A first embodiment of the restraining device generally includes a ring-shaped member, a pull-string, and a locking mechanism. This embodiment may be permanently or removably attached to an article of footwear. In this embodiment, the pull-string is configured to be inserted into the top portion of the ring-shaped member, such that pulling the pull-string tightens the top portion of the ring-shaped member around the leg/ankle/foot of the wearer. Once the ring-shaped member is tightened, the locking mechanism locks the pull-string in place, thereby inhibiting the removal of the article of footwear from the wearer, regardless of whether the footwear is tight or loose on the wearer's foot.

[0007] A second embodiment of the restraining device generally includes a ring-shaped member, tightening means, and a locking mechanism. This embodiment may also be permanently or removably attached to an article of footwear. In this embodiment, the tightening means and optionally the locking mechanism are configured to be hidden beneath or as part of the ring-shaped member, such that tightening means and locking mechanism are hidden from view of the footwear

while in use.

[0008] A third embodiment of the restraining device generally includes a restraining strap, a locking mechanism, and optionally a comfort pad. This embodiment may be permanently or removably attached to an article of footwear. In this embodiment, the restraining strap is configured to surround the leg/ankle/foot of the wearer. Once the restraining strap is tightened, the locking mechanism locks the restraining strap in place, thereby inhibiting the removal of the article of footwear from the wearer, regardless of whether the footwear's fastening means is fastened or not. The fourth embodiment of the restraining device utilizes a similar configuration as the third embodiment, but instead uses a different location for the locking mechanism when secured to the footwear.

[0009] A unique locking mechanism depicted in the third and fourth embodiments of the restraining device is also provided. This locking mechanism generally includes a strap insert and exit slot, a flexible retainer, a pair of pins, a pin holder for each pin, a push mechanism, and a locking plate. The interaction of these components in combination with the restraining strap creates a secure restraining device for inhibiting the removal of the footwear to which the restraining device is attached.

[0010] While these four exemplary embodiments are depicted for purposes of understanding the invention and concepts disclosed herein, other embodiments utilizing the same concepts are further contemplated.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] These and other features and advantages of the present invention will become appreciated, as the same becomes better understood with reference to the specification, claims and drawings herein:

[0012] FIG. 1 is a front perspective view of a first embodiment of a restraining device for inhibiting the removal of footwear secured to footwear.

[0013] FIG. 2 is a side view of the restraining device of FIG. 1 secured to footwear.

[0014] FIG. 3 is a rear view of the restraining device of FIG. 1 secured to footwear.

[0015] FIG. 4 is a top view of the restraining device of FIG. 1 secured to footwear.

[0016] FIG. 5 is a perspective view of the restraining device of FIG. 1.

[0017] FIG. 6 is a rear view of a locking mechanism of the restraining device of FIG. 1.

[0018] FIG. 7 is a top view of the locking mechanism of FIG. 6.

[0019] FIG. 8 is a perspective view of a second embodiment of a restraining device for inhibiting the removal of footwear secured to footwear.

[0020] FIG. 9 is a side view of the restraining device of FIG. 8 secured to footwear.

[0021] FIG. 10 is a top view of the restraining device of FIG. 8 secured to footwear.

[0022] FIG. 11 is a rear view of the restraining device of FIG. 8 secured to footwear having a hidden locking mechanism.

[0023] FIG. 12 is a perspective view of the restraining device of FIG. 8 having the hidden locking mechanism.

[0024] FIG. 13 is a side perspective view of a third embodiment of a restraining device for inhibiting the removal of footwear in an unlocked configuration.

[0025] FIG. 14 is a side perspective view of the restraining device of FIG. 13 in a locked configuration.

[0026] FIG. 15 is a rear perspective view of a fourth embodiment of a restraining device for inhibiting the removal of footwear secured to footwear.

[0027] FIG. 16 is a vertical cross-section of the locking mechanism of the restraining device of

FIG. 15.

[0028] FIG. 17 is a horizontal cross-section of the locking mechanism of the restraining device of FIG. 15.

[0029] FIG. 18 is a perspective view of the internal mechanisms of the locking mechanism of the restraining device of FIG. 15.

[0030] FIG. 19 is a top view of the internal mechanisms of the locking mechanism of the restraining device of FIG. 15 in a locked configuration.

[0031] FIG. 20 is a top view of the internal mechanisms of the locking mechanism of the restraining device of FIG. 15 in an unlocked configuration.

[0032] FIG. 21 is a perspective view of the locking plate of the locking mechanism of the restraining device of FIG. 15.

[0033] FIG. 22 is a top view of the inset area of the locking plate of the locking mechanism of the restraining device of FIG. 15.

DETAILED DESCRIPTION OF THE INVENTION

[0034] The invention now will be described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

[0035] It will be understood that when an element is referred to as being “on” another element, it can be directly on the other element or intervening elements may be present there between. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0036] It will be understood that, although the terms first, second, third etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section.

[0037] As used herein, the singular forms “a,” “an,” and “the,” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” “includes” and/or “including,” and “have” and/or “having,” when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof.

[0038] Furthermore, relative terms, such as “lower” or “bottom,” and “upper” or “top,” and “inner” or “outer,” may be used herein to describe one element's relationship to another elements as illustrated in the Figures. It will be understood that relative terms are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures. Unless otherwise defined, all terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0039] A restraining device configured to prevent footwear from being removed from the foot of a wearer is provided. The restraining device will prevent footwear from being removed regardless of the type of fastening means used to secure the footwear to the foot of a wearer and whether the fastening means are fastened. FIGS. 1-7 provide a first embodiment of the invention and concepts

disclosed. FIGS. 8-12 provide a second embodiment of the invention and concepts disclosed. FIGS. 13-14 provide a third embodiment of the invention and concepts disclosed. FIGS. 15-22 provide a fourth embodiment of the invention and concepts disclosed herein, having the same locking mechanism as the third embodiment. While these four exemplary embodiments are depicted for purposes of understanding the invention and concepts disclosed herein, other embodiments utilizing the same concepts are further contemplated.

[0040] FIGS. 1-7 depict the first embodiment of the restraining device **100**. FIGS. 1-4 show the restraining device **100** attached to an article of footwear **10**. The footwear **10** has an upper **11**, a lower **12**, a fastening means **13**, a tongue **14**, and a collar **15**. For purposes of this disclosure, lower is defined as the sole of the footwear, and upper is defined as the portion of the body of the footwear that houses the foot and includes the foot opening of the footwear. While footwear **10** having these elements is utilized in connection with the first embodiment, any type of footwear **10** may be utilized with the restraining device **100** without departing from the concepts disclosed herein. Generally, for all footwear, when a wearer inserts their foot into the footwear **10**, the wearer's foot is inserted through the collar **15** and is enclosed by the tongue **14**, upper **11**, and lower **12**. The fastening means **13** fasten the footwear **10** to the wearer's foot, such that while the fastening means **13** are secured, the footwear **10** remains secured around the wearer's foot.

Typically, the fastening means **13** are easily fastened or loosened such that the wearer can easily put on or take off the footwear **10** at their liking. For example, footwear **10** uses fastening means **13** such as shoelaces or hook-and-loop fasteners that are easily secured but similarly easily removed. [0041] While removing footwear **10** at a wearer's leisure is acceptable for most wearers, this is not the case for others, such as children and/or adults with special needs. The restraining device **100** is configured to secure the footwear **10** on these other wearers such that the footwear **10** cannot be removed by simply loosening the fastening means **13**.

[0042] In the first embodiment, the restraining device **100** has a ring-shaped member **110**, a pull-string **120**, and a locking mechanism **130**. The restraining device **100** is placed around the collar **15** and on top of the tongue **14** of the footwear **10**. The ring-shaped member **110** of the restraining device **100** has a top portion **112** and a bottom portion **114**, defined by the location of each portion when positioned on the footwear **10**. For illustrative purposes, if the ring-shaped member **110** was placed flat, the top portion **112** is the component defining the inner circle of the ring-shaped member **110**, and the bottom portion **114** is the component defining the outer circle of the ring-shaped member **110**. The ring-shaped member **110** may be made of any flexible material.

[0043] FIG. 5 shows the restraining device **100** not secured to footwear **10** and without the locking mechanism **130**. As shown in FIG. 5, the top portion **112** of the ring-shaped member **110** has a pull-string **120** placed throughout the top portion **112**. A portion of the pull-string **120** extrudes from the top portion **112** of the ring-shaped member **110** such that the pull-string can be pulled, thereby reducing the diameter of the inner circle of the ring-shaped member **110**. The lower portion **114** of the ring-shaped member **110** is configured to attach to the article of footwear **10**.

[0044] The restraining device **100** may be removably attached or permanently attached to the footwear **10**. For example, the lower portion **114** may be permanently sewn or glued to the upper **11** of the footwear **10**, or alternatively may be removably adhered to the upper **11**. While the exemplary embodiments show the lower portion **114** attached to the upper **11** of the footwear **10**, other embodiments may be configured so the lower portion **114** is capable of being attached to other pieces of the footwear **10**, such as the lower **12**, fastening means **13**, and/or the tongue **14**. Any attachment means providing the permanent or removable attachment of the restraining device **100** to the footwear **10** may be utilized without departing from the concepts disclosed herein.

[0045] As shown in FIGS. 1-4, once the lower portion **114** of the ring-shaped member **110** is attached to the footwear **10**, the upper portion **112** of the ring-shaped member **110** sits near or above the collar **15** of the footwear **10**. Once a wearer's foot is inserted through the collar **15** and into the footwear **10**, the wearer's foot, ankle, or lower leg will be surrounded and covered by the

top portion **112** of the ring-shaped member **110**. The wearer then pulls the pull-string **120** to tighten the top portion **112** of the ring-shaped member **110** around the wearer's foot/ankle/lower leg. The pull-string **120** is then locked into place by the locking mechanism **130**.

[0046] This system keeps the footwear **10** on the wearer's foot regardless of whether the fastening means **13** of the footwear **10** is fastened, as the restraining device **100** is tightened around the wearer's foot/ankle/lower leg. Importantly, the portion of the wearer's leg located below the top portion **112** of the restraining device **100** must be larger than the central opening of the ring-shaped member **110** when the top portion **112** is tightened via the pull-string **120**. Thus, even if the fastening means **13** of the footwear **10** is loosened, the wearer cannot remove the footwear **10** from their foot, as the size of the wearer's leg beneath the top portion **112** of the restraining device **100** is larger than the opening of the restraining device **100**. The restraining device **100** is particularly effective if the top portion **112** of the restraining device **100** is placed above the wearer's ankle, as the wearer's ankle will provide additional area that cannot be removed from the opening of the restraining device **100** absent loosening the pull-string **120**. The restraining device **100**, once tightened, will also inhibit the removal of the footwear **10** when the footwear **10** is under external forces. For example, the restraining device **100** will inhibit the removal of the footwear **10** if a wearer were to try to kick off the footwear **10** or use their hands to pull off the footwear **10**.

[0047] FIGS. 6-7 depicts a first exemplary embodiment of the locking mechanism **130** of the restraining device **100**. In this exemplary embodiment, the locking mechanism **130** is a "k-lock". This locking mechanism **130** is configured to accept the pull-string **120** and lock the pull-string **120** in place. In this exemplary embodiment, the pull-string **120** is inserted through the pull-string insert **132** and exits through the pull-string exit **134**. Once inserted, the sideways v portion of the "k-lock" locks the pull-string **120** in place, thus locking the restraining device **100** in place. Once the pull-string **120** is locked into place via the locking mechanism **130**, the restraining device **100** effectively inhibits the removal of the footwear **10**, regardless of whether the fastening means **13** of the footwear **10** are fastened or not. While a "k-lock" is presented in this exemplary embodiment shown in FIGS. 1-7, any other locking mechanisms may be similarly utilized without departing from the concepts disclosed herein, such as the locking mechanism **500** depicted in FIGS. 13-22.

[0048] FIGS. 8-12 provides a second embodiment of the invention and concepts disclosed. As shown in FIG. 12, this embodiment may utilize a different tightening means than the prior embodiment, as it does not have a tightening means external to the top portion **212** of the ring-shaped member **200**. Instead, unlike the pull-string **120** of FIGS. 1-7, this embodiment may keep the tightening means enclosed in the ring-shaped member **210** of the restraining device **200**. Numerous means of tightening the top portion **212** around the foot/ankle/lower leg of the wearer may be utilized in this embodiment, including the pull-string from the prior embodiment. For example, a chord may be placed inside the top portion **212** and connect to the locking mechanism **230**, such that the chord is not visible in the restraining device **200**. Another example is the fourth embodiment depicted in FIG. 15. In another embodiment, a tightening means of bunching the material at the top portion **212** of the ring-shaped member **200** may similarly inhibit the removal of the footwear **20**.

[0049] As shown in FIGS. 10-12, the locking mechanism **230** and tightening means may be hidden and not visible when attached to the footwear **20**. The general concepts disclosed in the first embodiment of tightening the top portion **212** of the ring-shaped member **210** to inhibit the removal of the footwear **20** is generally the same for this second embodiment. As in the first embodiment, the ring-shaped member **210** of the restraining device **200** is placed over the collar **25** and tongue **24** of the footwear **20** and attaches to the footwear **20** at preferably the upper **21** of the footwear **20**. Alternatively, the restraining device **200** may attach at the lower **22** of the footwear **20** or at any other portion of the footwear **20**.

[0050] Once the wearer inserts their foot into the footwear **20** through the restraining device **200**, the top portion **212** is tightened using the tightening means and locked into place using the locking

mechanism **230**. Thus, just as with the first embodiment, the restraining device **200** inhibits the removal of the footwear **20** regardless of whether the fastening means **23** of the footwear **20** are fastened. However, in this embodiment, the tightening means are hidden inside the top portion **212** of the ring-shaped member **210** and inside the locking mechanism **230**, such that the tightening means are not visible.

[0051] Furthermore, the locking mechanism **230** may also be hidden underneath or behind the ring-shaped member **210**, as shown in FIGS. **10-11**, such that the wearer does not see the locking mechanism **230** while the restraining device **200** is attached to the footwear **20**. This is particularly effective at inhibiting particular groups of wearers, such as children or adults with special needs, from locating and unlocking the locking mechanism **230**. In this embodiment, a caretaker of this group of wearers may still access the locking mechanism **230** while locking the restraining device **200**, then cover the locking mechanism **230** with the ring-shaped member **210**, thereby concealing the locking member **230**. FIGS. **10-12** display this embodiment of the restraining device **200**, where the ring-shaped member **210** is covering the locking mechanism **230** such that it is inaccessible to the previously discussed group of wearers.

[0052] FIGS. **13-14** depict a third embodiment of the restraining device **300** secured to an article of footwear **30**. FIG. **13** depicts the third embodiment in an unlocked configuration, while FIG. **14** depicts the third embodiment in a locked configuration. In general, the third embodiment of the restraining device **300** includes a restraining strap **310** and a locking mechanism **500**. Optionally, the third embodiment may also include a comfort pad **312**. For ease of reference, the footwear **30** in this third embodiment includes an upper **31**, a lower **32**, a fastening means **33**, and a tongue **34**.

[0053] In this third embodiment, the restraining strap **310** is preferably removably attached to the footwear **30**. To attach the restraining strap **310** to the footwear **30**, two openings are included in the fabric of the inside of the upper **31** on each side of the footwear **30**, thereby creating an entry and exit for the restraining strap **310**. Two apertures may also be made in the footwear tongue **34**, forming one opening on each side of the footwear tongue **34**. This creates two open passages through the footwear tongue **34**, one on the left side of the footwear tongue **34** and the other on the right side of the footwear tongue **34**, through which the restraining strap **310** passes.

[0054] Thus, as shown in FIG. **13**, when the restraining device is in its unlocked position, the ends of the restraining strap **310** will exit through the front of the footwear tongue **34** and be configured to accept the locking mechanism **500**. Prior to the locking mechanism **500** being locked, the restraining strap **310** is in a loose configuration, such that the restraining strap **310** is not inhibiting the removal of the footwear **30**.

[0055] As shown in FIG. **14**, once the locking mechanism **500** is placed onto the footwear **30** and locked in place, the restraining strap **310** can be locked around the wearer's foot, ankle, or lower leg, thereby inhibiting the removal of the footwear **30**, even if the fastening means **33** is loosened or undone. Thus, when the restraining device **300** is in its unlocked position, the footwear **30** may be put on and taken off as with any footwear **30**. Yet, when the restraining device **300** is in its locked position, the footwear **30** may not be put on or taken off, even if the footwear's fastening means **33** is loosened.

[0056] In this third embodiment, the concepts disclosed are configured to permit the locking mechanism **500** to be placed on the front of the footwear **30**, specifically contacting the footwear tongue **34**, as depicted in FIGS. **13-14**. The restraining straps **310** are thus placed on the inside of the footwear **30** such that the restraining device **300** and footwear **30** are not detached from each other. In these embodiments, the use of a comfort pad **312** attached to the restraining strap **310** at the heel of the footwear **30** is preferable, especially if the material of the restraining strap **310** is uncomfortable. The inclusion of the comfort pad **312** provides comfort to the wearer's heel and achilleas tendon while wearing the footwear **30** having the restraining device **300**. Further, the comfort pad **312** may further prevent any damage, such as blisters that may occur due to the tight fit of the restraining device **300** around the wearer's foot/ankle/lower leg. In some embodiments,

the comfort pad **312** may not be necessary, especially should the material of the restraining strap **310** provide sufficient comfort to the wearer.

[0057] The restraining strap **310** may be manufactured of any suitable material. For example, the restraining strap **310** may be manufactured out of nylon, polyester, rubber, cotton, leather, neoprene, or other synthetic rubbers, fabrics, or elastics. Preferably, the restraining strap **310** will be manufactured out of a non-elastic material, such as leather. However, elastic compounds may be used as well so long as their elasticity is not so great that the restraining strap **310** is unable to inhibit the removal of the footwear **30** when the restraining device **300** is in a locked configuration. The comfort pad **312** may be manufactured out of any comfortable material, such as cotton, polyester, foam, memory foam, or similar compounds. Any material that provides greater comfort on the heel and achilleas of the wearer than the material used for the restraining strap **310** may be utilized for the comfort pad **312**.

[0058] FIG. **15** depicts the fourth embodiment of the restraining device **400** attached to an article of footwear **40**. In this embodiment, the configuration of the restraining device **400** and the footwear **40** is very similar to the third embodiment, except the locking mechanism **500** is configured to be placed on the back of the footwear **40**. In this fourth embodiment, the footwear includes an upper **41** and a lower **42**. While this slip-on style of footwear is depicted for this embodiment, any suitable footwear may be utilized.

[0059] The restraining device **400** includes a restraining strap **410** and a locking mechanism **500**. In this embodiment, the restraining strap **410** is configured to wrap around the top portion of the upper **41**, with the locking mechanism **500** on the heel of the upper **41**. As with the previous embodiments, the restraining device **400** may be configured to be removably attached to the footwear **40** or may be permanently attached as part of the footwear **40**. In this embodiment, as well as the third embodiment, the unique structure of the locking mechanism **500** accompanied by the restraining strap **410** provides locking capabilities such that the restraining device **400** will inhibit the removal of the footwear **40** and the locking mechanism **500** will be difficult for particular groups of wearers to unlock.

[0060] FIGS. **16-22** depict the unique structure of the locking mechanism **500** used for the third and fourth exemplary embodiments. As shown in FIGS. **16-18**, the locking mechanism **500** includes a strap insert slot **502**, a strap exit slot **504**, a flexible retainer **506** which may include a retainer block **507**, two pins **508**, a pin holder **509** for each pin **508**, a push mechanism **510**, and a locking plate **512**. Each of these components are housed inside the locking mechanism **500** such that only the restraining straps **410** and the push-end of the push mechanism **510** are viewable, as depicted in FIGS. **13-15**.

[0061] Inside the locking mechanism **500** is the flexible retainer **506**. The flexible retainer **506** is configured to be of a flexible material, such that the flexible retainer **506** may be pushed and pulled into ovals of different shape. Abutting the outside of the two vertexes of the flexible retainer **506** are each push mechanism **510**. The push mechanisms **510** are configured to be slightly convex on the side outside of the locking mechanism **500** and are concave on the inside of the locking mechanism **500** such that the outside of the ends of the flexible retainer **506** sit securely within concave section of the push mechanism **510**. Abutting the outside of the two co-vertexes of the flexible retainer **506** are the retainer blocks **507**. The retainer blocks **507** are configured to contact the restraining straps **410** when the locking mechanism **500** is in its locked configuration, while not contacting the restraining straps **410** when the locking mechanism **500** is in its unlocked configuration. In alternative embodiments, the retainer blocks **507** may not be utilized, in which case the flexible retainer **506** will instead be configured to interact with the restraining straps **410**.

[0062] On the inside of the flexible retainer **506** sit the pins **508**, the pin holders **509**, and the locking plate **512**. On the inside of the two vertexes of the flexible retainer **506** are the two pin holders **509**, each at one vertex of the flexible retainer **506**. In some embodiments, the pin holders **509** are constructed as a single piece with the flexible retainer **510**. In other embodiments, the pin

holders **509** are attached to the flexible retainer **506**. In other embodiments, such as the embodiment depicted by FIG. **16**, the pin holders **509** are constructed as a single piece with the push mechanism **510**. In these embodiments, a slot will be positioned between the push mechanism **510** and the pin holders **509**, the slot configured such that the flexible retainer **506** sits in the slot of the push mechanism **510** and pin holders **509**. In each of these embodiments, pushing the push mechanism **510** causes the flexible retainer **506** to push the pin holder **509**, thereby moving the pins **508** inside the locking plate **512**.

[0063] Located in between the two pin holders **509** is the locking plate **512**. The locking plate **512**, depicted on its own in FIG. **21**, includes two inset areas **520** that are generally heart shaped. As shown in FIGS. **16** and **18**, the pins **508** are generally in an upside-down “U-shape,” similar to the shape of a staple. Each pin **508** is configured to have one end placed in its respective pin holder **510**, while the other end of each pin **508** sits within the inset area **520** of the locking plate **512**. To function, the end of the pin **508** placed in the pin holder **509** may be capable of rotating, such that the end of the pin **508** inside the inset area **520** is able to shift side-to-side.

[0064] FIGS. **19-20** depict the functionality of the locking mechanism **500**. FIG. **19** depicts the locking mechanism **500** in its locked configuration, while FIG. **20** depicts the locking mechanism **500** in its unlocked configuration. While in the unlocked position, each end of the restraining strap **410** is inserted into the locking mechanism **500** through each strap insert slot **502** and exit the locking mechanism **500** through each strap exit slot **504**. Thus, when the locking mechanism **500** is in its unlocked configuration, the restraining straps **410** may be freely pulled in and out of the locking mechanism **500**, thereby tightening and loosening the restraining device **500**. When the locking mechanism **500** is in its locked configuration, the restraining straps **410** are locked in place in the locking mechanism **500**, thereby inhibiting the removal of the footwear **40** to which the restraining device **400** is attached.

[0065] Preferably, the flexible retainer **506** will be configured to be in its unlocked shape shown in the unlocked configuration of FIG. **20** when at rest. As such, the vertexes of the flexible retainer **506** will always be in contact with the push mechanisms **510**. The push mechanisms **510** preferably have an edge configured to prevent the push mechanism **510** from being fully pushed out of the locking mechanism **500**, such that in the unlocked configuration, the convex end and a portion of the push mechanism **510** are outside the locking mechanism **500**. In the unlocked configuration, the flexible retainer **506** (and retainer block **507** if utilized) will not be in contact with the restraining strap **410**. In the locked configuration, the flexible retainer **506** (and retainer block **507** if utilized) will be pressed against the restraining strap **410** with sufficient force to prevent the restraining strap **410** from being loosened or tightened.

[0066] The locking and unlocking of the locking mechanism **500** are initiated by pushing the push mechanism **510**. As shown in FIG. **20**, when the locking mechanism **500** is in its unlocked configuration, the pins **508** are positioned in the base level **521** of the locking plate **512**. Locking the locking mechanism **500**, as shown in FIG. **19**, is caused by pushing in on the push mechanisms **510**, thereby causing the flexible retainer **506** to be squeezed from an ovular shape into a more circular shape. This in turn causes the vertexes of the flexible retainer **506** to push on the pin blocks **509**, while simultaneously causing the co-vertexes of the flexible retainer **506** (or retainer block **507** if utilized) to push against the restraining straps **410**. Pushing in the pin blocks **509** through the above process causes the pins **508** to shift between the base level **521** of the inset area **520** and the lock level **524** of the inset area **520**. Once the pins **508** are pushed into the lock level **524** of the locking plate **512**, the locking mechanism **500** is shifted into its locked configuration, as shown in FIG. **19**. Thus, in this position, the pins **508** are locked into the lock levels **524** of the locking plate **512** such that the flexible retainer **506** (or retainer block **507** if utilized) is pushed against the restraining straps **410**, thereby locking the restraining straps **410** in place.

[0067] To unlock the locking mechanism **500**, the push mechanisms **510** are again pushed in, which in turn shifts the pins **508** from the lock level **524** of the inset area **520** back to the base level **521** of

the inset area **520**. This in turn allows the flexible retainer **506** to return to its more ovular shape, where the flexible retainer **506** (or retainer block **507** if utilized) is no longer contacting the restraining straps **410**.

[0068] FIGS. **21-22** depict the locking plate **512** and inset areas **520** of the locking plate **512**. The unique configuration of the locking plate **512** and inset areas **520**, accompanied by the pressure caused by the shape of the flexible retainer **506**, allows the pins **508** to shift between the base level **521** and lock level **524**, thereby locking and unlocking the locking mechanism **500**. Each heart shaped inset area **520** of the locking plate **512** has at least one of the following five levels; a base level **521**, a ramp up **522**, a plateau **523**, a lock level **524**, and an intermediate level **525**.

[0069] Starting in the unlocked position, the pin **508** will sit in the inset area **520** at the base level **521** closest to the pin holder **509**. When the push mechanism **510** is pushed in, the pin **508** will be pushed from the base level **521** up the ramp up **522**, past the plateau **523**, and into the intermediate level **525**. Due to the force of the flexible retainer **506** attempting to move back into its unlocked, more ovular shape, the pin **508** will automatically be pulled back towards the pin holder **509**, which causes the pin to drop down to the lock level **524** of the inset area **520**. Due to the concave shape of the center area **526** of the locking plate **512**, the pin **508** will be locked into the lock level **524** and will be unable to exit the lock level **524** without external pressure on the push mechanism **510**.

Thus, the location of the pin **508** inside the lock level **524** of the inset area **520** accompanied by the pressure caused by the flexible retainer **506** attempting to return to its unlocked configuration will keep the locking mechanism **500** locked, in turn causing the restraining device **400** to inhibit the removal of the footwear **40** to which it is attached.

[0070] To return the locking mechanism **500** to its unlocked position, the push mechanism **510** is pressed to move the pin **508** inward out of the lock level **524** and into the second base level **521**. Once the pin **508** is pushed into this base level **521**, the force of the flexible retainer **506** trying to return to its unlocked configuration will force the pin **508** to move up the ramp up **522**, past the plateau **523**, and back into the first base level **521**, back to its unlocked configuration. Thus, the unique interplay between the configuration of the inset area **520** of the locking plate **512** and the force caused by the flexible retainer **506** enable this locking mechanism **500** to lock and unlock the restraining device **400**. While specific locking mechanisms have been described, one skilled in the art could substitute known locking mechanisms and remain within the scope of the invention to provide a footwear removal restraint.

[0071] Each of the discussed exemplary embodiments, as well as any other embodiments protected by the concepts disclosed herein, may be configured to be attached to any piece of footwear as a separate restraining device, or integrated into an article of footwear, such that the restraining device is part of the footwear. For example, the third embodiment depicted by FIG. **15** may be configured to be a part of the footwear **40** such that the restraining strap **410** and/or locking mechanism **500** cannot be removed from the footwear **40** or be configured to be removable from the footwear **40** such that the restraining device **400** can be attached to any article of footwear **40**. Preferably, the restraining device is configured to be removably attachable to any article of footwear **10**, **20**, **30**, **40**.

[0072] Exemplary embodiments of the present invention are described herein with reference to idealized embodiments of the present invention. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments of the present invention should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

Claims

1. An article of footwear for enclosing a user's foot, the article of footwear comprising: a lower comprising a sole of the footwear; an upper disposed above the lower, the lower and the upper defining a body for housing the user's foot, the upper having an opening; the opening comprising a collar and a tongue, the collar and the tongue configured such that the user's foot can be inserted therethrough and surrounded by the body; a restraining strap configured to surround the user's foot at the opening; a fastener configured to receive the restraining strap with a portion of the restraining strap extending through the fastener to engage the restraining strap to hold the restraining strap in a fixed position and a release for disengaging the restraining strap from the fixed position; and wherein a portion of the restraining strap extends away from the opening.
2. The article of footwear of claim 1 further comprising a ring-shaped member surrounding the opening, the ring-shaped member having an upper portion and a lower portion.
3. The article of footwear of claim 2 wherein the restraining strap is a pull-string disposed around the upper portion of the ring-shaped member, the pull-string comprising a loop.
4. The article of footwear of claim 3 wherein the fastener is a locking mechanism having a "v" portion for locking the pull-string in place.
5. The article of footwear of claim 4 wherein the fastener is located on a rear portion of the footwear.
6. The article of footwear of claim 1 wherein the restraining strap is woven into the collar and the tongue.
7. The article of footwear of claim 6 wherein the restraining strap extends through the tongue.
8. The article of footwear of claim 6 wherein the fastener is disposed on the tongue with terminal ends of the restraining strap extending through the fastener.
9. The article of footwear of claim 6 further comprising a comfort pad disposed adjacent the collar opposite the tongue.
10. The article of footwear of claim 9 wherein the restraining strap extends through the comfort pad.
11. The article of footwear of claim 1 wherein the restraining strap extends around the outside of the upper.
12. The article of claim 11 wherein the fastener is disposed on a rear portion of the footwear.
13. An article of footwear for enclosing a user's foot, the article of footwear comprising: a lower comprising a sole of the footwear; an upper disposed above the lower, the lower and the upper defining a body for housing the user's foot, the upper having an opening; the opening comprising a collar and a tongue, the collar and the tongue configured such that the user's foot can be inserted therethrough and surrounded by the body; a restraining strap configured to surround the user's foot at the opening; a fastener comprising a locking pin and a push release mechanism; and the fastener is configured to engage the restraining strap with a portion of the restraining strap extending through the fastener; the fastener includes a release to disengage the fastener from the restraining strap; and wherein a portion of the restraining strap extends away from the opening.
14. The article of footwear of claim 13, further comprising a ring-shaped member surrounding the opening, the ring-shaped member having an upper portion and a lower portion.
15. The article of footwear of claim 14 wherein the restraining strap is a pull-string disposed around the upper portion of the ring-shaped member, the pull-string comprising a loop.
16. The article of footwear of claim 13 wherein the restraining strap is woven into the collar and the tongue.
17. The article of footwear of claim 16 wherein the restraining strap extends through the tongue.
18. The article of footwear of claim 16 wherein the fastener is disposed on the tongue with terminal ends of the restraining strap extending through the fastener.
19. The article of footwear of claim 16 further comprising a comfort pad disposed adjacent the collar opposite the tongue.

20. The article of footwear of claim 19 wherein the restraining strap extends through the comfort pad.
